SIMATS ENGINEERING
Slot: A, Course Code: ITA0513, Course Name: Computer Vision

Duration: 60 mins Marks: 20 Assessment Tool 1: Short Answer Test (High) Assessment Tool Weightage: 40%

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)

Q. No.	Questions
1	A factory uses cameras to inspect products moving on a conveyor belt. Explain how Computer Vision can identify defective products automatically.
2	A facial recognition system is affected by changes in pose and illumination. Explain which Computer Vision features can help improve recognition performance.
3	A traffic monitoring system needs to detect vehicles, recognize traffic signs, and estimate distances. Explain how different levels of Computer Vision are involved.
4	A scanned document contains blurred characters and unwanted noise. Explain how digital image fundamentals can be used to improve the document before further processing.
5	A smartphone camera produces different image quality under indoor and outdoor conditions. Explain the factors involved in image sensing and acquisition.
6	A surveillance camera captures a moving person as a blurred object. Identify possible causes related to image acquisition and suggest suitable improvements.
7	A digital image contains very few pixels and important object details are missing. Explain how the sampling process affects the image quality.
8	An image contains smooth intensity variations, but after digitization it shows noticeable intensity steps. Explain how quantization causes this effect.
9	A remote sensing system must capture very small objects on the Earth's surface. Explain how the choice of sampling resolution affects the ability to detect these objects.
10	A grayscale image is stored using 2 bits per pixel instead of 8 bits per pixel. Explain how this change affects the number of intensity levels and image appearance.

Code of Conduct:

*I **Midhuna A (192521302)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student
Midhuna A
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RUBRICS FOR CALCULATION

Criteria	Weigh tage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts

Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

COMPUTER VISION

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1. computer vision can use cameras and AI inspect products conveyor belt

* camera capture product image
* image processing identifiers cracks, scratches, wrong size, missing parts.
* AI compares product with normal product
* If a detects found the system automically rejects remaves the product.
* The system can store the defect type and counter quality control analysis.

Example: A camera detects a cracked bottle sends signal remove conveyor belt.

Features can improve the facial recognition are

* SIFT / SURF - Handle changes in scale, rotation, pose
* HOG - captures facial shape and edges
* LBP - Helps changes illumination and texture
* Deep CNN features - Learn robust facial features and improve recognition under different pose and lightning.

* Deep CNN features provides strong performance for pose and illumination.

3. Different levels of computer vision are used:

* Low level: Image processing remove noise and improve image quality
* Mid level: Detects vehicle traffic signals using the edges and features
* High level: Recognizes traffic signals understand the
* 3D / Depth analysis: Estimates distance of vehicles from camera.

4. Digital image fundamentals improve the scanned docum
Noise removal: use filters to remove unwanted noise
Image sharpening: mark blured characters clearer
Contrast enhancement: Improve difference between text background
Thresholding: convert the image into clear black an white text.
Segmentation: seperate characters or text from the background.

5. Image quality depends on:

Lightning: Indoor light usually lower than the outdoor
Illumination Intensity: Affects brightness and exposure
Camera' Sensor: Sensor size and sensitity affect quality.
Lens - Focus and aperture control amount of light.

* Exposure time: Longer exposure can cause blur
* Image resolution: Higher resolution gives more details

6. Possible cause

* Slow shutter speed / exposure time
* poor lighting
* camera or object movement
* Incorrect focus

Improvements:

* use a faster shutter speed
* Increase lighting
* use autofocus
* Increase.

7. Effect sampling on image quality:

Sampling converts a continous image into pixels.
Low sampling rate → fewer pixels
important object details may be lost
image may appear pixelated or blurry
Higher sampling rate → more pixels and better deta

8. Quantization converts continuous intensity values fixed le

* It fewer intensity levels are used, smooth changes

became visible steps.

* This effect called quantization error / banding
* more intensity levels give smoother images.

Sampling resolution determines smallest object can be detected

Low resolution - small object may disappear merge surroundin

High resolution → more pixels represent object

Therefore high spatial sampling resolution need detect very small object.

10 2 bit vs 8 bit Grayscale image

2 bits / pixel : $2^2 = 4$ intensity levels

8 bits / pixel : $2^8 = 256$ intensity levels

2 bit image fewer shades rough, posterized less detailed.